

Cryptographic Techniques for Privacy and Security in the Age of A(G)I

Ghada Almashaqbeh

University of Connecticut

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Before AGI

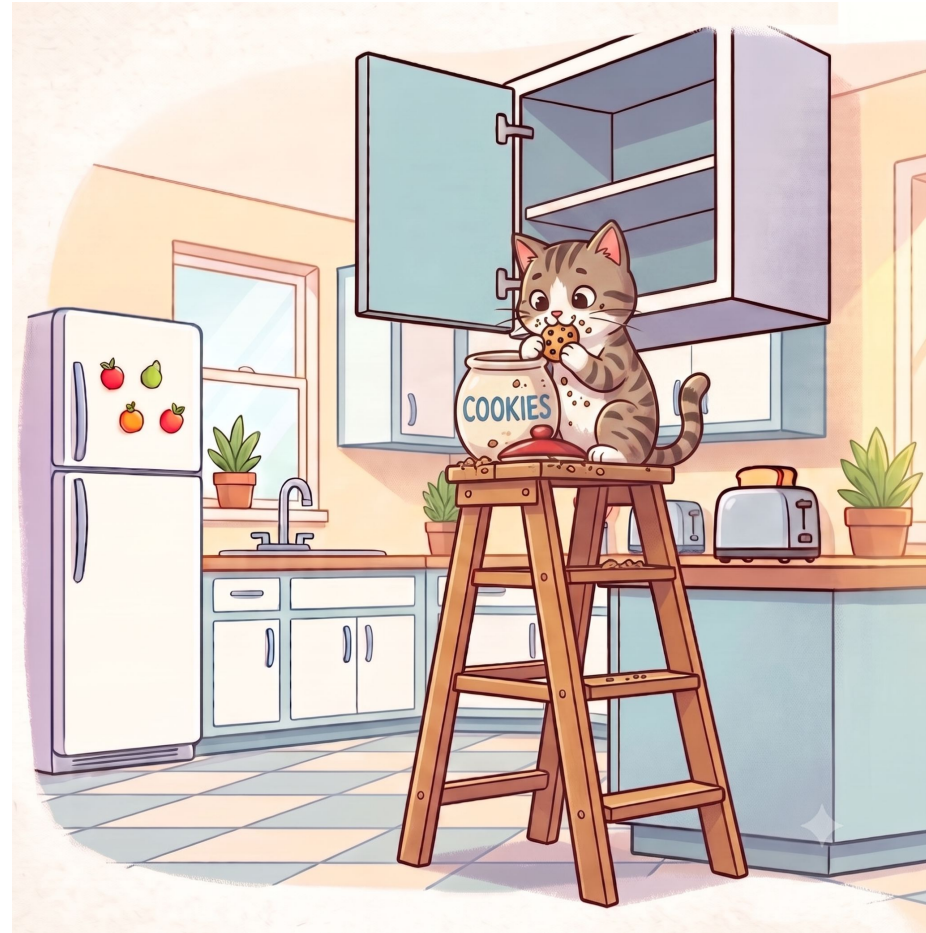
My system runs this specific protocol and behaves in this particular way.



* AI generated

After AGI

My system evolves over time;
it observes my (and others')
data and learns!

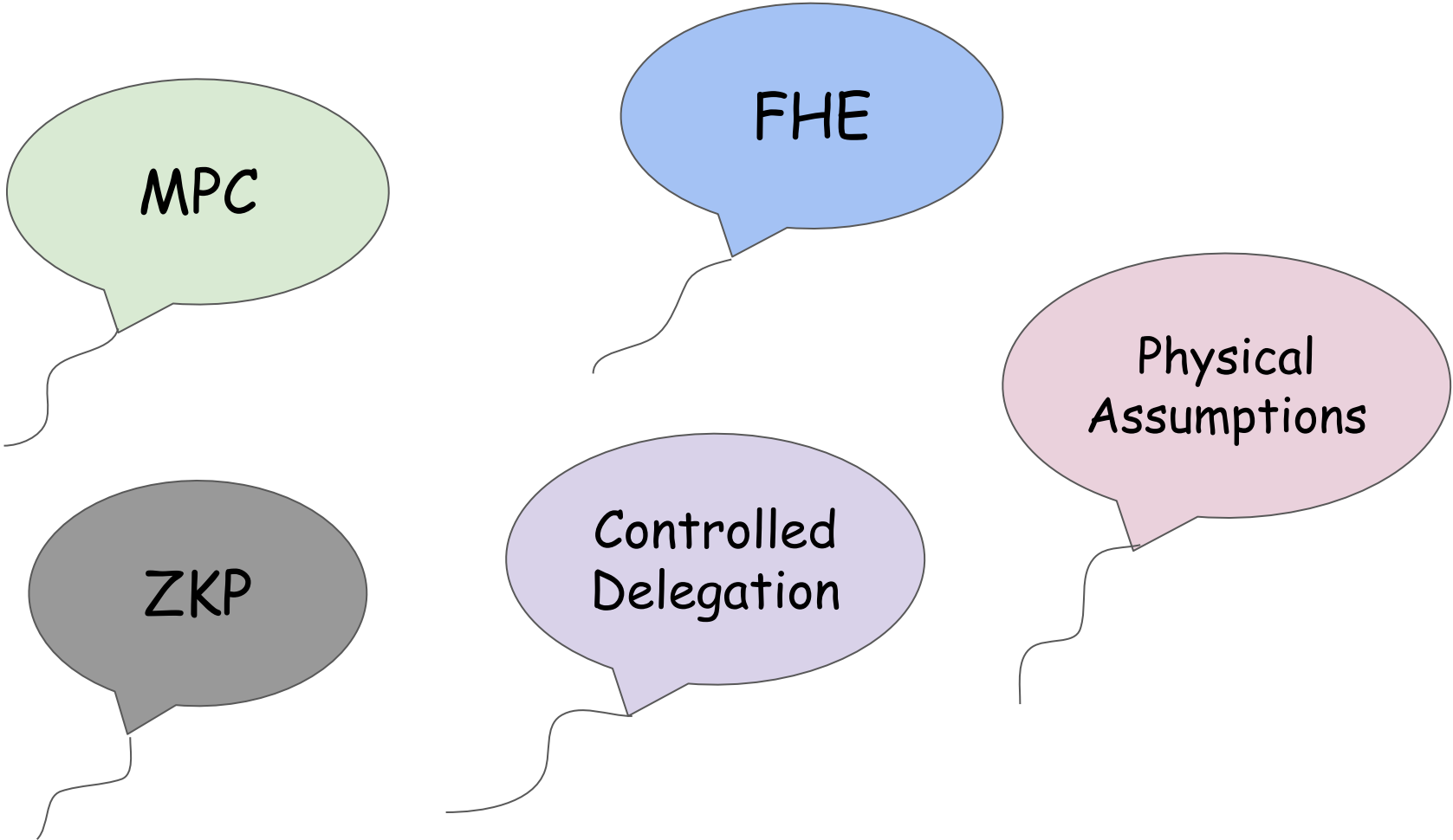


* AI generated

Security and Privacy of *Heterogeneous* Collaborative Systems

- Handle participants/systems of different capabilities
 - Different trust levels, resource availability, actor types (conventional vs. agentic), etc.
- How to account for these discrepancies in:
 - Private computing.
 - Trust distribution.
 - Capability delegation.
 - ...

Ingredients ...



MPC

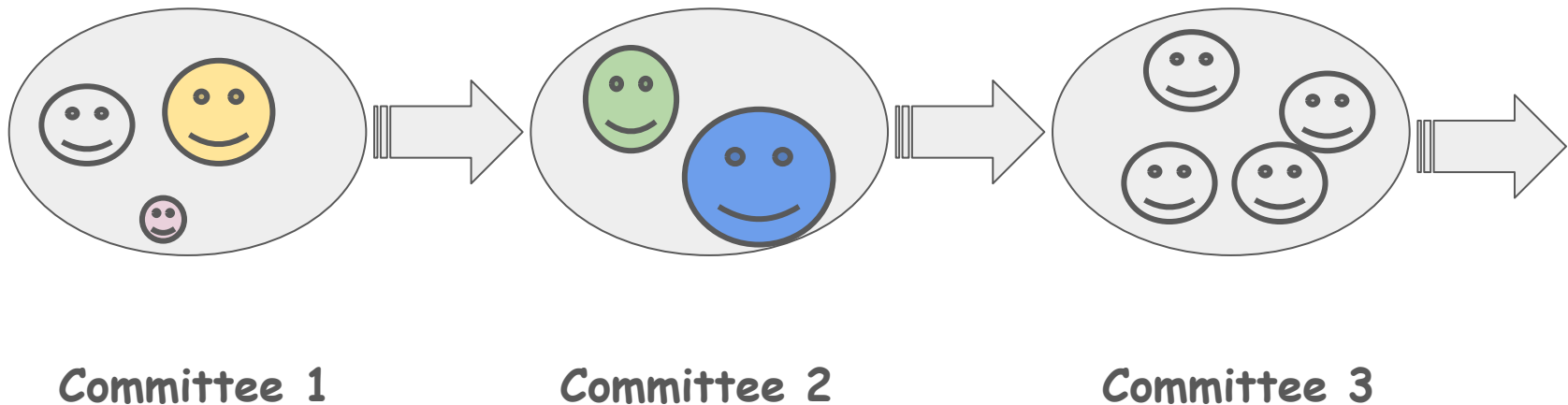
FHE

Physical
Assumptions

ZKP

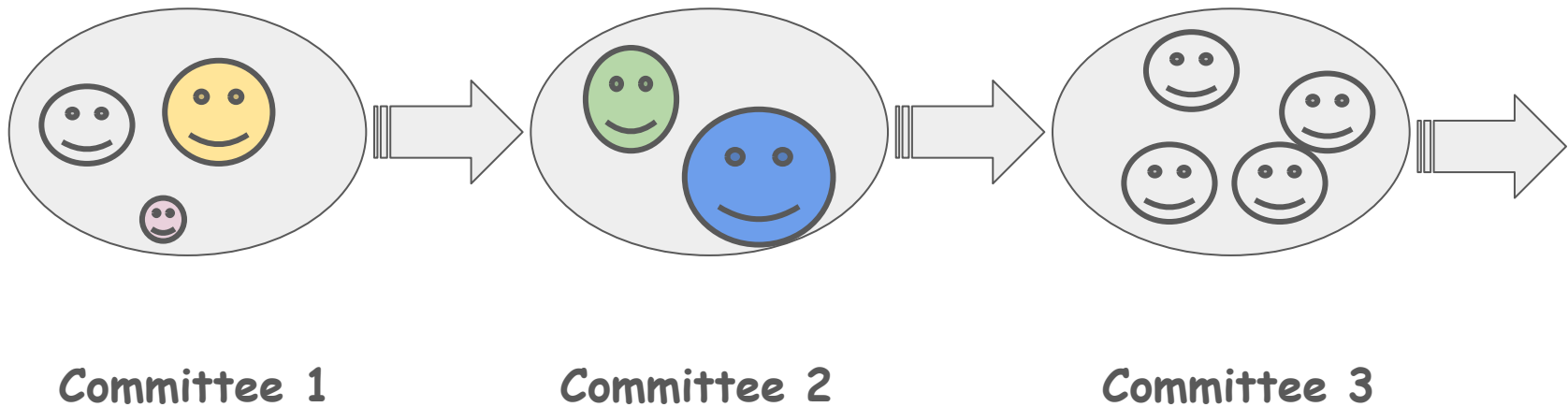
Controlled
Delegation

Weighted (Fluid) MPC



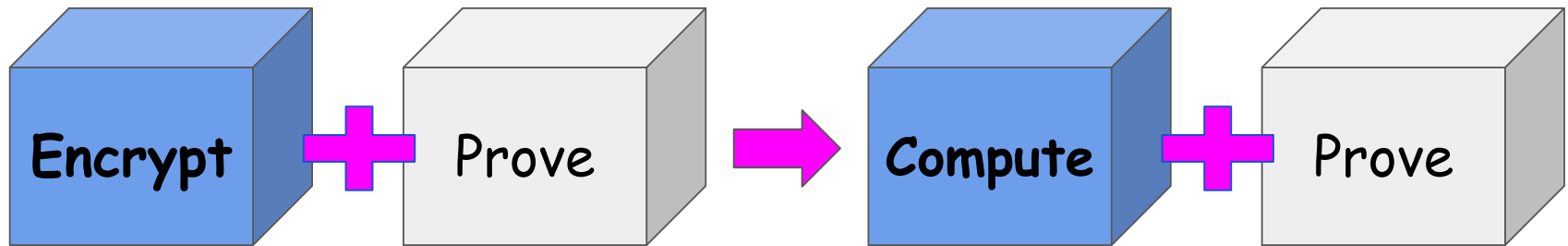
- Participants have weights reflected on the shares they receive $\Rightarrow$ VWRSS scheme.
- Participants can come and go based on resource availability $\Rightarrow$ committee-based MPC with handover.

Weighted (Fluid) MPC



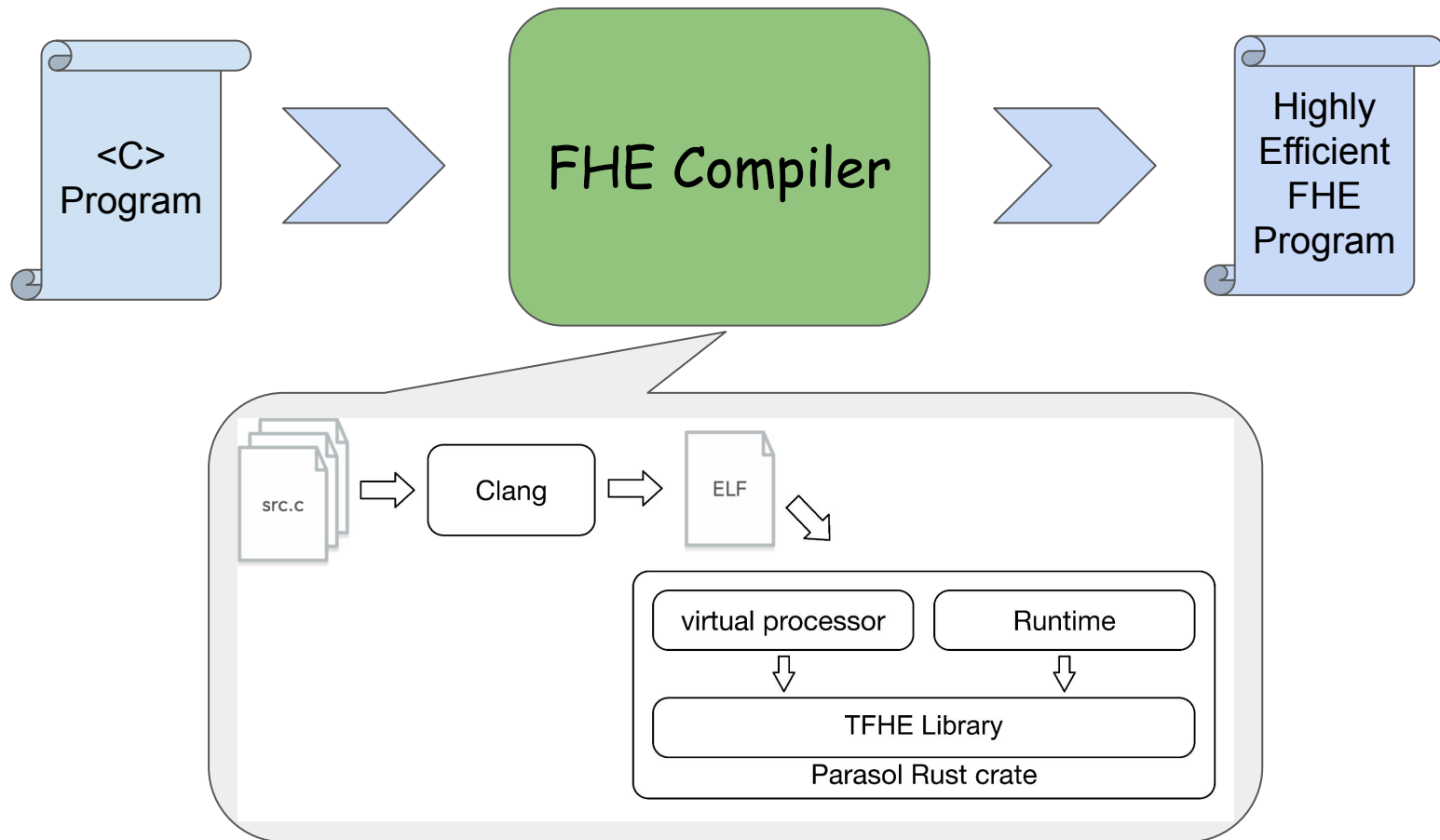
- This “diversity” would enable: capturing adversary evolution, enforcing human in the loop, ...
- How does it behave in practice?

Fully Homomorphic Encryption (FHE)



- I/O Verifiability?!
 - Input: satisfy specific properties + ciphertext well-formedness $\Rightarrow$ ZKP
 - Output: computation correctness $\Rightarrow$ ZKP or consensus

Efficiency and Usability $\Rightarrow$ Parasol Compiler



Efficiency and Usability



- How about more?
 - Scheme switching.
 - Weighted threshold FHE.

Controlled Delegation



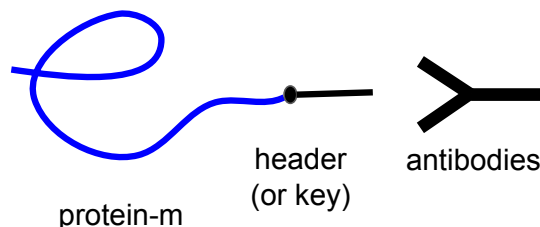
- I trust you but not so much :)
 - Timed, anonymous, revocable and policy-compliant delegation.
- Accountability?! *Do not blame me, blame the AI agent...*
- More functionalities?

Polymer-based Cryptography

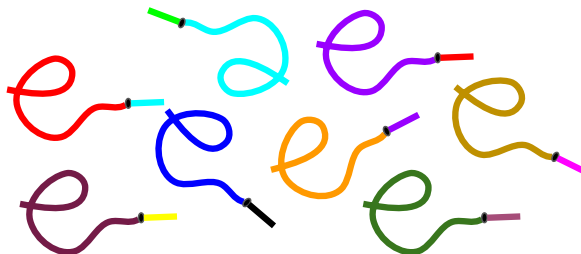
- Anchoring AGI to physical assumptions.
 - *Bro, it is not personal; it is just how it works!*

Protein-based consumable memory gadgets.

Synthesize m



Mix with decoy proteins

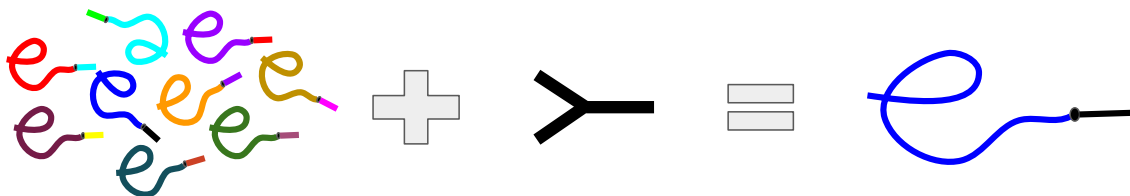


Polymer-based Cryptography

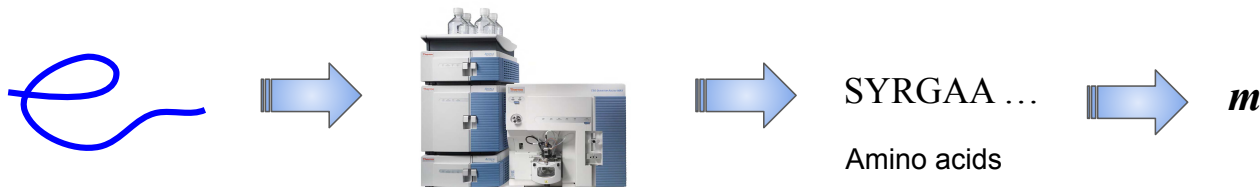
- Anchoring AGI to physical assumptions.

Protein-based consumable memory gadgets.

To retrieve m , first purify

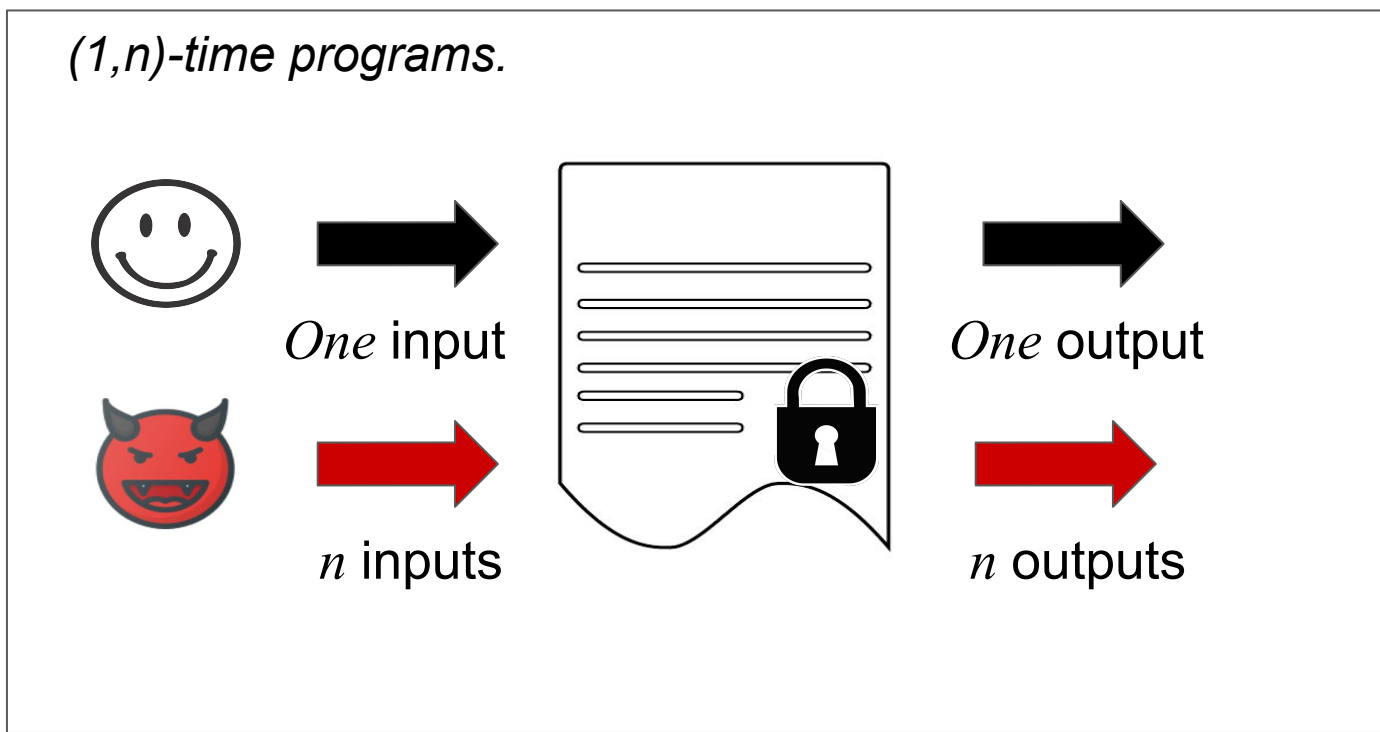


then read the sequence



Polymer-based Cryptography

- Anchoring AGI to physical assumptions.



Wait, how to define security?



Thank you!

Questions?